



LinkedIn Post Generator — AI Agent

Module 20 Assignment | LangChain + Google Gemini



Project Overview

An AI-powered agent that generates professional LinkedIn posts based on user input.

Users can control:

- Topic (e.g. "AI in Healthcare", "Remote Work Productivity")
 - Language (English, Bengali, Spanish, French, Arabic, Hindi, Portuguese, German, Mandarin)
 - Tone (Professional, Inspirational, Storytelling, Data-driven, Conversational, Thought Leader)
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Tech Stack

- Python
 - Flask
 - LangChain
 - Google Gemini API
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Project Structure

```
linkedin_agent/ |— agent.py |— app.py |— requirements.txt |— README.md |— templates/ |—  
index.html |— static/ |— css/style.css |— js/main.js
```



How to Run

1. Install dependencies

```
pip install langchain langchain-google-genai google-genai flask python-dotenv
```

2. Add API Key

Create a .env file: GEMINI_API_KEY=your-gemini-api-key

3. Run project

```
python app.py
```

4. Open browser

<http://localhost:5000>

Features

- AI-powered LinkedIn post generation
 - Multi-language support (9 languages)
 - Multiple tone selection
 - Flask REST API
 - Modern UI (HTML + CSS + JS)
 - Copy & regenerate functionality
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Workflow

User Input → Flask API → Gemini AI → Prompt Engineering → LinkedIn Post Output

Module 20 Assignment

Course: AI Agents with LangChain

Student: ASHIQ RAHMAN CHOWDHURY

Notes

- Uses latest Gemini API
 - Keep API key secure (.env file)
 - Do not upload secrets to GitHub
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Future Improvements

- LinkedIn auto-posting system
- Post scheduling feature
- Analytics dashboard
- Advanced prompt optimization